# Nisarg Parikh

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## Education

**University of Massachusetts, Amherst** *Amherst, MA*

Doctor of Philosophy in Computer Science (3.85/4.0 GPA) *Sep 2024 - Expected May 2030*

- **Advisor:** Andrew Lan
- **Courses:** Reinforcement Learning, Advanced Algorithms, Distributed and Operating Systems, Machine Learning, Hot topics in Software Engineering Research.

**Northeastern University, Khoury College of Computer Science** *Boston, MA*

Master's of Science in Computer Science (3.83/4.0 GPA) *Sep 2022 - May 2024*

- **Courses:** Reinforcement Learning, Procedural Content Generation, Empirical Research Methods, Natural Language Processing, Machine Learning, Computer Systems, Algorithms, Programming Design Paradigms.

## Work Experience

**Adobe** *Bangalore, India*

Research Intern *May 2025 - Aug 2025*

- Developed a confidence-calibration objective for Transformer language models within a preference optimization framework.
- Investigated calibration-aware language modeling to improve the reliability of model confidence estimates during LLM evaluation.
- Proposed an automated evaluation approach leveraging calibrated language models to reduce manual effort in product development.

**Elemental Cognition** *New York (Remote)*

Data Science/ Machine Learning Intern *Jun 2023 - Aug 2023*

- Developed a Python-based repository to query and transform BigQuery data into structured tabular outputs for analysis and presentation.
- Wrote 10+ features to customize results fetched from Big Query, added an option to group keywords together.
- Evaluated proprietary NLP tools across 50+ prompts, identifying 10+ failure modes and informing improvements to model behavior.
- Contributed to a contrastive NLP task within an internal vector search tool.

## Research

**CATTO: Balancing Preferences and Confidence in Language Models**

**Nisarg Parikh**, Kunjal Panchal, Ananya Sai, Pannaga Shivaswamy, Andrew Lan *Spring 2025 - Present*

*Under review @ AAAI 2027.* CATTO is a calibration-aware token-level objective that regularizes confidence during preference optimization by minimizing a differentiable surrogate of an upper bound on Expected Calibration Error.

**LookAlike: Consistent Distractor Generation in Math MCQs**

**Nisarg Parikh**, Alexander Scarlatos, Nigel Fernandez, Andrew Lan *October 2024 - Present*

*Published @ BEA 2025.* Studying the benefit of iterative optimization with preference based RL approaches such as DPO and supervised approaches to consistency in distractor generation.

**Thinking Forward: Memory-Efficient Federated Finetuning of Language Models**

Kunjal Panchal, **Nisarg Parikh**, Sunav Choudhary, Yuriy Brun, and Hui Guan *Spring 2024*

*Published @ NeurIPS, 2024.* Spry is a federated learning algorithm that enables finetuning LLMs using Forward-mode Auto Differentiation.

**Flow: Per-Instance Personalized Federated Learning**

Kunjal Panchal, Sunav Choudhary, **Nisarg Parikh**, Lijun Zhang, and Hui Guan *Spring 2023*

*Published @ NeurIPS, 2023.* Flow addresses the challenge of statistical heterogeneity through creating dynamically personalized models.

## Skills

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|---|---|
| **Programming** | Python (PyTorch, Tensorflow, scikit-learn, Numpy, Pandas, matplotlib, HuggingFace), Java, C/C++. |
| **Machine Learning** | Natural Language Processing, Large Language Models (LLMs), Federated Learning, Preference Optimization, DPO, In-Context Learning, Hypothesis Testing. |

## Projects

**Mario with GFlowNets**

Github - https://github.com/namelessCrusader/Mario-GFlowNet/ *Feb 2024 - May 2024*

- Implemented Flow Matching and Trajectory Balance objectives for Generative Flow Networks (GFlowNets) to procedurally generate Super Mario Bros. levels.
- Developed a custom level-generation environment and represented generated levels using the Video Game Level Corpus (VGLC) format.

**Stochastic Wave Function Collapse**

Github - https://github.com/namelessCrusader/Stochastic-Wave-Function-Collapse/ *Jan 2024 - Feb 2024*

- Implemented a stochastic variation of the Wave Function Collapse algorithm to procedurally generate Super Mario Bros levels.
- Utilised ASCII representation of Mario levels to learn compositional details and constraints with the Video Game Level Corpus dataset.
- Procedurally generated 100 levels of 200 x 14 tiles.

## Leadership/Volunteering

Current **Peer Reviewer**, NeurIPS ’26 (Gold Reviewer), ICML ’26, AAAI ’27